Adaptive Saturated ADRC of a Piezoelectric Drug-Delivery Actuator with Cubic Damping and Dahl Hysteresis

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***Abstract*— This letter develops an adaptive saturated ADRC scheme for position tracking of a piezoelectric actuator (PEA) for liquid drug delivery. The actuator combines cubic viscous damping with Dahl hysteresis, its effective input gain is unknown, and the driving voltage saturates. An extended-state observer, driven by the applied voltage, reconstructs the unmeasured velocity and the matched plant perturbation from position measurements alone, while a projection-based adaptive law keeps the nominal input gain inside a known positive interval. In contrast with cancellation-based designs, the cubic damping is retained in the closed loop: its dissipativity confines the trajectories, so the operating envelope is derived, and it absorbs the damping mismatch in the observer, so no lower bound on the observer bandwidth is required. The tracking and observer errors are proven semiglobally uniformly ultimately bounded, and simulations confirm the analysis under active saturation.**

***Index Terms*— Active disturbance rejection control, adaptive control, state observer, input saturation, piezoelectric actuators (PEA).**

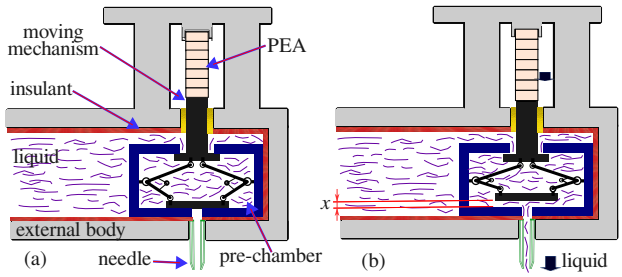


Fig. 1: Piezoelectric drug-delivery device. The voltage u applied to the PEA drives a rhombus mechanism, immersed in the liquid, that opens two gaps in an antagonist way: (a) liquid is admitted into the pre-chamber; (b) it is injected towards the needle through the second gap, whose displacement x sets the delivered dose. The motion is affected by the Dahl hysteresis of the PEA and by the cubic damping of the liquid, and x is regulated from position measurements under a saturated u.

I. Introduction

Piezoelectric actuators (PEAs) provide sub-micrometric positioning resolution making them suitable for applications requiring extreme precision [1]. In liquid-medium positioning however, the actuator dynamics are affected by nonlinear drag, which can be represented by a cubic velocity-dependent damping term. Moreover, PEAs exhibit voltage-dependent hysteresis that may significantly degrade tracking accuracy if it is not considered in the control design [2]. The paper context is the control of a PEA devoted to precisely deliver liquid drug (Figure 1). The system exhibits nonlinear damping term and strong hysteresis behavior.

Hysteresis compensation in PEAs is usually addressed through either feedforward or feedback strategies. Feedforward methods rely on the inverse of an identified hysteresis model, such as Bouc-Wen, Prandtl-Ishlinskii, or Preisach operators [3], [4], cascaded with the plant [5], an approach that continues to advance through rate-dependent hysteresis modeling [6]. Although simple to implement, their performance can deteriorate in the presence of unmodeled disturbances. Feedback methods are more robust and either compensate hysteresis before a nominal controller or include the hysteresis dynamics directly in the observer/controller design. Recent observer-based designs estimate the hysteresis state and reject it within the feedback loop, using cascade or high-gain hysteresis observers combined with robust or sliding-mode control for PEAs with Bouc-Wen or modified-Dahl models [7]–[9], typically establishing Lyapunov-based convergence of the estimation and tracking errors. In this context, the Dahl model is attractive because it captures non-symmetric, friction-like hysteresis in piezoelectric materials. It has also been combined with output-feedback active disturbance rejection control [10], [11] for PEAs under linear viscous damping [12].

This letter considers a more demanding setting, where Dahl hysteresis is combined with cubic viscous damping, an unknown input gain, and voltage saturation. Related issues such as ADRC under actuator saturation [13], input-gain uncertainty, and disturbance-observer-based robust control [14] have been studied in broader contexts; however, their combination for the PEA considered here has not been addressed

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Supplementary material available at github.com/gfloresc/adrc-pea-drug-delivery.

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to our knowledge. Figure 1 shows the delivery system considered. The controller uses position measurements only, and the contribution is threefold. First, the position and velocity envelopes are established from plant data: the saturated input, the bounded Dahl state and the cubic dissipation confine the trajectories to a computable region. Second, the bounded-rate condition that ADRC normally assumes is here derived, from the plant parameters and the exogenous force alone. Third, the unknown input gain is estimated by a projection-based law whose estimation error is part of the stability argument, and the saturation residual is treated explicitly. The dissipativity of the cubic damping is exploited twice, and no lower bound on the observer bandwidth is imposed. The resulting property is semiglobal uniform ultimate boundedness.

II. Modeling and Problem Statement

Figure 1 displays a schematic diagram of the delivery system. When a voltage u is applied to the PEA, it expands and pushes a rhombus mechanism that opens two gaps in an antagonist way: one draws high-pressure liquid into the pre-filled pre-chamber, and the other injects it precisely towards the needle. The exact amount of the injected drug is determined by the displacement x of this second gap. The model that relates displacement x with the voltage is:

$$\Sigma : \begin{cases} m_p\ddot{x} = d_p u - h - s_p F - c_p(\dot{x})^3 - k_p x & \text{(1a)} \\ h = \gamma \mathrm{sgn}(\dot{x}) z & \text{(1b)} \\ \dot{z} = -\alpha \dot{x} - \beta \left|\dot{x}\right| z & \text{(1c)} \end{cases}$$

where h is an internal state variable depending on the hysteresis state z and the velocity $\dot{x}$, and F is an external unknown force applied to the actuator and to the mechanism. Here m_p, c_p and k_p are the effective mass, damping coefficient and stiffness, d_p is the voltage-to-force homogenization coefficient, s_p the compliance coefficient, and γ, α, β define the hysteresis size and shape; all are positive. The damping term is cubic because the moving mechanism is immersed in pressurized liquid. The sole control input is the voltage u and the output of interest is the displacement:

$$y = x. \tag{2}$$

A. Problem statement

The PEA operates in a liquid medium, with the voltage u as its only control input and the displacement $y = x$ as its measured output. The velocity $\dot{x}$, the environmental force $F(t)$, the Dahl state z, and the hysteresis force h are unavailable for feedback. The problem is as follows.

Problem 1. *For system* (1a)-(2)*, design an observer-based voltage input u such that $y = x$ tracks $y_d(t)$ in the presence of cubic viscous damping, Dahl hysteresis, an unknown effective input gain and an unknown environmental force, while the applied voltage is limited to $|u| \le u_{\max}$. The controller must use only the displacement measurement, together with the reference signals y_d, $\dot{y}_d$, and $\ddot{y}_d$, and must guarantee ultimate boundedness of the tracking and observer errors without postulating bounds on the plant states.*

III. Main Result

With $x_1 = x$ and $x_2 = \dot{x}$, system (1a) becomes

$$\begin{aligned} \dot{x}_1 &= x_2, \\ \dot{x}_2 &= bu + f_d(x_2) + \xi(x_1, x_2, z, t), \end{aligned} \tag{3}$$

where $b := d_p/m_p$ is the effective input gain and $f_d(x_2) := -\frac{c_p}{m_p} x_2^3$ is the nonlinear viscous damping term. Since $h = \gamma \,\mathrm{sgn}(x_2) z$ in (1b) is discontinuous at $x_2 = 0$, for the theoretical analysis, consistently with the numerical implementation of the supplementary material, we replace $\mathrm{sgn}(x_2)$ by

$$\sigma_\varepsilon(x_2) := \tanh\left(\frac{x_2}{\varepsilon_s}\right), \qquad \varepsilon_s > 0, \tag{4}$$

and define the matched lumped perturbation as

$$\xi(x_1, x_2, z, t) := -\frac{k_p}{m_p} x_1 - \frac{\gamma}{m_p} \sigma_\varepsilon(x_2) z - \frac{s_p}{m_p} F(t), \tag{5}$$

which collects the stiffness, regularized Dahl hysteresis, and environmental-force terms. In contrast, the cubic damping term f_d is kept explicit in the closed loop, so that its dissipative contribution can be used in the Lyapunov analysis (Lemma 2). The replacement (4) is needed because the analysis below uses $\dot{\xi}$, which fails to exist where $\gamma \,\mathrm{sgn}(x_2) z$ jumps at a sign-changing zero crossing of x_2. The modelling error it induces is quantified by

$$\left| \frac{\gamma}{m_p} \big(\mathrm{sgn}(x_2) - \sigma_\varepsilon(x_2)\big) z \right| \le \frac{2\gamma \bar{z}}{m_p} \, e^{-2|x_2|/\varepsilon_s}, \tag{6}$$

which decays exponentially outside a band of width $O(\varepsilon_s)$ around $x_2 = 0$. The bounds of Proposition 1 hold for either model, as they only use $|\mathrm{sgn}| = |\sigma_\varepsilon| \le 1$.

Assumption 1. *The effective input gain $b = d_p/m_p$ is an unknown constant that belongs to a known interval, $b \in [b_{\min}, b_{\max}]$, with $0 < b_{\min} \le b_{\max}$. Consequently $d_p = m_p b$ is unknown as well. All remaining plant parameters are known.*

Assumption 2. *The desired position $y_d(t)$ is a bounded C^2 trajectory, and there exist known positive constants $\bar{y}_0$, $\bar{y}_1$ and $\bar{y}_2$ such that $|y_d(t)| \le \bar{y}_0$, $|\dot{y}_d(t)| \le \bar{y}_1$ and $|\ddot{y}_d(t)| \le \bar{y}_2$ for all $t \ge 0$.*

Assumption 3. *The environmental force $F : [0, \infty) \to \mathbb{R}$ is continuously differentiable and there exist known constants $\bar{F}, \bar{F}_1 > 0$ such that $|F(t)| \le \bar{F}$ and $|\dot{F}(t)| \le \bar{F}_1$ for all $t \ge 0$.*

The analysis rests on three properties of the plant alone, none of which involves the controller.

Lemma 1. *Let z be the solution of the Dahl hysteresis subsystem*

$$\dot{z} = -\alpha x_2 - \beta |x_2| z. \tag{7}$$

Then, for every initial condition $z(0)$, the hysteresis state is bounded. In particular, $|z(t)| \le \bar{z}$, $\forall t \ge 0$, where

$$\bar{z} := \max\left\{ |z(0)|, \frac{\alpha}{\beta} \right\}. \tag{8}$$

Proof. See Section S-I of the supplementary material. □

Proposition 1 (A priori boundedness of the plant)**.** *Let Assumptions 1 and 3 hold and let the applied voltage be the saturated command* (14)*, so that* $|u(t)| \le u_{\max}$ *for all* $t \ge 0$*. Define the total exogenous force bound*

$$\bar{w} := m_p b_{\max} u_{\max} + \gamma \bar{z} + s_p \bar{F}, \tag{9}$$

and, for free analysis parameters $0 < \kappa < \sqrt{k_p/m_p}$ *and* $x_s > 0$ *that do not enter the design and may be optimized to tighten the bounds, let* $\psi(x_1) := x_1/\sqrt{1+(x_1/x_s)^2}$ *(so that* x_s *sets its saturation level,* $|\psi| \le x_s$*),* $\rho(x_1) := x_1\psi(x_1)$*,* $c_0 := \kappa^2 m_p x_s^2$ *and*

$$\Psi(s) := c_p s^4 - \kappa c_p x_s s^3 - \kappa m_p s^2 - \bar{w} s. \tag{10}$$

Let $\bar{s}$ *be the largest positive root of* $\Psi(s) = \kappa \bar{w} x_s$*, let* $\Phi := \max_{s\in[0,\bar{s}]}\big(-\Psi(s)\big)$ *and* $X_1 := \rho^{-1}\big((\Phi + \kappa\bar{w}x_s)/(\kappa k_p)\big)$*. Then, for every initial condition of the regularized model* (3)–(5)*, the plant trajectories satisfy, for all* $t \ge 0$*,*

$$|x_1(t)| \le \bar{x}_1, \qquad |x_2(t)| \le \bar{x}_2, \tag{11}$$

where, *with* $\bar{V}_p := \frac{3}{4} m_p \max\{|x_2(0)|, \bar{s}\}^2 + \frac{1}{2} k_p \max\{|x_1(0)|, X_1\}^2 + c_0$*,*

$$\begin{aligned} \bar{x}_1 &:= \sqrt{\frac{2(\bar{V}_p + c_0)}{k_p}}, \\ \bar{x}_2 &:= \max\Big\{|x_2(0)|,\ \big(\tfrac{\bar{w}+k_p\bar{x}_1}{c_p}\big)^{1/3}\Big\}. \end{aligned} \tag{12}$$

The constants $\bar{x}_1, \bar{x}_2$ *are independent of the controller, of the observer and of* $\hat{b}$*.*

Proof. See Section S-II of the supplementary material. □

Remark 1. *For fixed plant parameters and exogenous-signal bounds, the constants of Proposition 1 depend on the plant initial condition only through* $|x_1(0)|$*,* $|x_2(0)|$ *and* $\bar{z} = \max\{|z(0)|, \alpha/\beta\}$*. Hence, for any compact set* $\mathcal{X}_0 \subset \mathbb{R}^3$ *of admissible plant initial conditions, uniform upper bounds on these three quantities yield constants* $\bar{x}_1, \bar{x}_2$*, and every constant built from them, in particular* L_d*,* C_g *and* $\bar{\zeta}_1$*, that are uniform over* $\mathcal{X}_0$*.*

Lemma 2 (Damping monotonicity)**.** *The map* $f_d : \mathbb{R} \to \mathbb{R}$*,* $f_d(v) = -\frac{c_p}{m_p} v^3$*, is monotonically non-increasing:* $(f_d(v) - f_d(w))(v - w) \le 0$ *for all* $v, w \in \mathbb{R}$*. Moreover, the following quantitative version holds:*

$$(f_d(v) - f_d(w))(v - w) \le -\frac{c_p}{4m_p}(v - w)^4, \qquad \forall\, v, w \in \mathbb{R}. \tag{13}$$

Proof. See Section S-I of the supplementary material. □

The scheme is developed next and summarized in Fig. 2: the measured position drives an extended-state observer whose estimates feed an integral tracking law, the command passes through a smooth saturation, and a projection-based law updates the nominal input gain.

A. Smooth saturated control

The driving voltage is limited by the amplifier rating $u_{\max}$. Therefore, the applied input is generated from the unsaturated command u_c through the smooth saturation

$$u = \mathrm{sat}_s(u_c) := u_{\max} \tanh\left(\frac{u_c}{u_{\max}}\right). \tag{14}$$

The saturation residual $\Delta u := \mathrm{sat}_s(u_c) - u_c$ satisfies the sector property $u_c\,\Delta u \le 0$, and the bound $|\Delta u| \le |u_c|$, $\forall\, u_c \in \mathbb{R}$. The function sat_s is smooth, odd, and bounded by $u_{\max}$; hence $|u| \le u_{\max}$ for all $t \ge 0$. Note that $\Delta u \ne 0$ even when the amplifier limit is not reached, because sat_s is a smooth compression rather than a hard clip.

B. Adaptive input-gain estimation

The nominal gain $\hat{b}(t)$ is generated online by

$$\dot{\hat{b}} = \gamma_b \,\mathrm{Proj}_{[b_{\min}, b_{\max}]}\big(\hat{b},\ \tau\big), \quad \tau := 2u(P_c\hat{\chi})_2, \quad \gamma_b > 0, \tag{15}$$

where u is the applied voltage (14), $\hat{\chi} := [e_1, \hat{e}_2, \eta]^\top$ collects the measurable tracking errors (33), and $P_c \succ 0$ is the constant matrix solving the Lyapunov equation (39) of the tracking dynamics, which depends on the design gains k_1, k_2, k_I and λ_I only and is computed offline. Every signal in τ is therefore available online. The projection is the standard scalar operator

$$\mathrm{Proj}_{[b_{\min}, b_{\max}]}(\hat{b}, \tau) := \begin{cases} 0, & \hat{b} = b_{\max},\ \tau > 0, \\ 0, & \hat{b} = b_{\min},\ \tau < 0, \\ \tau, & \text{otherwise}, \end{cases} \tag{16}$$

which renders $[b_{\min}, b_{\max}]$ invariant and satisfies, for every $\tau \in \mathbb{R}$ and every $b \in [b_{\min}, b_{\max}]$,

$$\tilde{b}\big[\mathrm{Proj}(\hat{b}, \tau) - \tau\big] \le 0, \qquad \tilde{b} := \hat{b} - b. \tag{17}$$

Equation (15) is thus a projected differential equation; since the constraint set is a closed convex interval and τ is locally Lipschitz in the remaining closed-loop states, it admits an absolutely continuous solution, and the inequalities involving $\hat{b}$ hold for almost every t.

Two consequences are used below. The estimate remains in $[b_{\min}, b_{\max}]$ for any $\hat{b}(0)$ in that interval, so $\hat{b} \ge b_{\min} > 0$ and the control law is always well defined; the estimation error obeys

$$|\tilde{b}(t)| \le \bar{b} := b_{\max} - b_{\min}, \qquad \forall t \ge 0, \tag{18}$$

regardless of γ_b. The proven role of the adaptive law is the Lyapunov cancellation in Step 3 of the proof of Theorem 1; convergence of $\hat{b}$ to b is not claimed and would require persistent excitation. The estimate is initialized at any $\hat{b}(0) \in [b_{\min}, b_{\max}]$.

C. Extended-state observer

Since only x_1 is measured, the velocity x_2 and an extended matched perturbation (defined below) are reconstructed through the extended-state observer

$$\begin{aligned} \dot{\hat{x}}_1 &= \hat{x}_2 + \ell_1(x_1 - \hat{x}_1), \\ \dot{\hat{x}}_2 &= \hat{b}u + f_d(\hat{x}_2^c) + \hat{\zeta} + \ell_2(x_1 - \hat{x}_1), \\ \dot{\hat{\zeta}} &= \ell_3(x_1 - \hat{x}_1), \end{aligned} \tag{19}$$

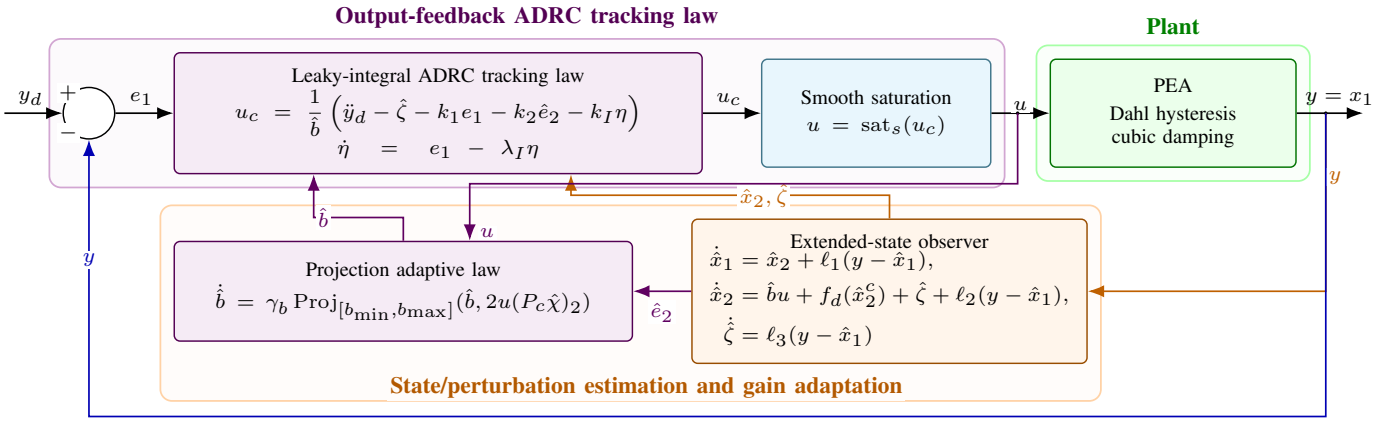


Fig. 2: Block diagram of the proposed adaptive saturated output-feedback ADRC scheme. The ESO, driven by the applied voltage u, reconstructs $\hat{x}_2$ and $\zeta = \xi$ from the measured position $y = x_1$; the integral state carries a leakage $\lambda_I > 0$.

where $u = \mathrm{sat}_s(u_c)$ is the applied voltage (14), $\ell_1, \ell_2, \ell_3 \in \mathbb{R}_{>0}$ are chosen, for a bandwidth $\omega_o > 0$ and fixed normalized gains $\bar{\ell}_1, \bar{\ell}_2, \bar{\ell}_3 > 0$, as

$$\ell_i = \omega_o^i \bar{\ell}_i, \qquad i = 1, 2, 3, \tag{20}$$

such that

$$s^3 + \ell_1 s^2 + \ell_2 s + \ell_3 \tag{21}$$

is Hurwitz, equivalently $s^3 + \bar{\ell}_1 s^2 + \bar{\ell}_2 s + \bar{\ell}_3$ Hurwitz, and

$$\hat{x}_2^c := \mathrm{sat}_{\bar{x}_2}(\hat{x}_2) \tag{22}$$

clamps the velocity estimate to the envelope $[-\bar{x}_2, \bar{x}_2]$ of Proposition 1 before it is used inside $f_d(\cdot)$; the estimate $\hat{x}_2$ fed to the controller is not clamped.

Driving the ESO with the applied voltage u, rather than with the unsaturated command u_c, allows the *extended matched perturbation* to be reduced to the plant terms alone,

$$\zeta(t) := \xi(x_1, x_2, z, t), \qquad \Delta u := u - u_c. \tag{23}$$

Neither the saturation residual nor the input-gain mismatch is lumped into ζ: the former is handled in the tracking analysis and the latter by the adaptive law (15). Indeed, using $b = \hat{b} - \tilde{b}$,

$$\hat{b}u + \zeta(t) - \tilde{b}\,u = \hat{b}u - \tilde{b}u + \xi = bu + \xi(t), \tag{24}$$

so that the true plant dynamics (3) are, exactly (not approximately),

$$\dot{x}_2 = \hat{b}u + f_d(x_2) + \zeta(t) - \tilde{b}\,u. \tag{25}$$

The gain mismatch appears here as the explicit term $-\tilde{b}u$, with the bounded regressor u, and is cancelled, up to the observer error, by the update (15).

Because $\zeta = \xi$ depends only on the plant state and on F, Proposition 1, Lemma 1 and Assumption 3 give at once

$$|\zeta(t)| \le \bar{\zeta}_0 := \frac{k_p \bar{x}_1 + \gamma \bar{z} + s_p \bar{F}}{m_p}, \tag{26}$$

so $|\zeta|$ is bounded by a constant built from the plant data. The rate of (23) can now be bounded from the same data.

Lemma 3 (Derived disturbance-rate bound)**.** *Under Assumption 3 and Proposition 1, the extended matched perturbation* (23) *is differentiable and* $|\dot{\zeta}(t)| \le \bar{\zeta}_1$ *for all* $t \ge 0$*, with*

$$\bar{\zeta}_1 := \frac{k_p}{m_p}\bar{x}_2 + \frac{\gamma}{m_p}\left(\frac{\bar{a}_2 \bar{z}}{\varepsilon_s} + \bar{z}_1\right) + \frac{s_p}{m_p}\bar{F}_1, \tag{27}$$

where $\bar{a}_2 := b_{\max} u_{\max} + \frac{c_p}{m_p}\bar{x}_2^3 + \bar{\zeta}_0$ *bounds* $|\dot{x}_2|$*,* $\bar{z}_1 := (\alpha + \beta\bar{z})\bar{x}_2$ *bounds* $|\dot{z}|$*, and* $\bar{\zeta}_0$ *is given by* (26)*.*

Proof. See Section S-I of the supplementary material. □

Remark 2. *The bound* (27) *is uniform over* $\mathcal{X}_0$ *and has the form* $\bar{\zeta}_1 = C_{\zeta,0} + C_{\zeta,1}/\varepsilon_s$*. Reducing the regularization width* ε_s *therefore improves the pointwise approximation* (6) *of the discontinuous Dahl term away from zero velocity, but enlarges the worst-case analytical bound.*

The damping mismatch entering the observer obeys two bounds, used jointly below.

Lemma 4 (Local Lipschitz bound)**.** *Under Proposition 1 and the clamp* (22)*,*

$$\left|f_d(\hat{x}_2^c) - f_d(x_2)\right| \le L_d\,|d| \le L_d|\tilde{x}_2|, \qquad L_d := 3\frac{c_p}{m_p}\bar{x}_2^2, \tag{28}$$

where $\tilde{x}_2 := \hat{x}_2 - x_2$ *and* $d := \hat{x}_2^c - x_2$*. Moreover* d *has the same sign as* $\tilde{x}_2$ *and* $|d| \le |\tilde{x}_2|$*, so, writing* $g := f_d(\hat{x}_2^c) - f_d(x_2)$*,*

$$g\,\tilde{x}_2 \le g\,d \le -\frac{c_p}{4m_p}d^4 \le 0. \tag{29}$$

Proof. See Section S-I of the supplementary material. □

Both bounds enter the estimate of Proposition 2 below, the first in the cross term and the second in the dissipative one; combining them is what makes that estimate hold for every $\omega_o > 0$.

Proposition 2 (Decoupled observer-error estimate)**.** *Let Assumptions 1–3 hold and let* $\hat{b}(0) \in [b_{\min}, b_{\max}]$*, so that* $|\tilde{b}| \le \bar{b}$ *by* (18)*. Let* $\varepsilon := \left[\tilde{x}_1,\ \tilde{x}_2/\omega_o,\ \tilde{\zeta}/\omega_o^2\right]^\top$ *with* $\tilde{\zeta} := \hat{\zeta} - \zeta$*, let* $\bar{A}$ *be the companion matrix of* $s^3 + \bar{\ell}_1 s^2 + \bar{\ell}_2 s + \bar{\ell}_3$ *(Hurwitz by* (20)*) and let* $\bar{P} \succ 0$ *solve* $\bar{A}^\top \bar{P} + \bar{P}\bar{A} = -I$*, which is fixed and independent of* ω_o*. Fix compact sets* $\mathcal{X}_0, \mathcal{C}_0$ *of*

plant and compensator initial conditions, let $\mathcal{F} := \{F \in C^1 : |F(t)| \le \bar{F},\ |\dot{F}(t)| \le \bar{F}_1\}$ *be the admissible force class, and set* $\varepsilon_{0,\max} := \sup_{\mathcal{X}_0\times\mathcal{C}_0\times\mathcal{F}} \|\varepsilon(0)\|$*, which accounts for the dependence of* $\tilde{\zeta}(0) = \hat{\zeta}(0) - \xi(x_1(0), x_2(0), z(0), F(0))$ *on* $F(0)$*. With* $\bar{E}_1 := [0,1,0]^\top$*,* $\bar{E}_o := [0,0,1]^\top$*, define*

$$\bar{c}_o := 2\|\bar{P}\bar{E}_o\|, \quad \bar{c}_b := 2\|\bar{P}\bar{E}_1\|\bar{b}\,u_{\max}, \quad C_g := \frac{m_p L_d^4}{2\,c_p\,\bar{P}_{22}\,\omega_o^4}, \tag{30}$$

and, in terms of these,

$$K_o := \frac{\bar{c}_o\bar{\zeta}_1}{\omega_o^2} + \frac{\bar{c}_b}{\omega_o}, \qquad D_\varepsilon := C_g + \frac{K_o^2}{\omega_o}. \tag{31}$$

Then, for every $\omega_o > 0$*, with no lower bound imposed on the observer bandwidth,*

$$\|\varepsilon(t)\|^2 \le \bar{\varepsilon}^2 := \frac{\lambda_{\max}(\bar{P})}{\lambda_{\min}(\bar{P})} \max\left\{\varepsilon_{0,\max}^2,\ \frac{4D_\varepsilon}{\omega_o}\right\} \tag{32}$$

for all $t \ge 0$ *and every initial condition in* $\mathcal{X}_0 \times \mathcal{C}_0$*, and* $\limsup_{t\to\infty} \|\varepsilon(t)\| \le \left[4\lambda_{\max}(\bar{P})D_\varepsilon/(\omega_o\lambda_{\min}(\bar{P}))\right]^{1/2}$.

Proof. See Section S-III of the supplementary material. □

D. Observer-based adaptive tracking controller

Let us define the measured position error, the estimated velocity error, and the integral state as

$$e_1 = x_1 - y_d, \qquad \hat{e}_2 = \hat{x}_2 - \dot{y}_d, \qquad \dot{\eta} = e_1 - \lambda_I\eta, \quad \lambda_I > 0. \tag{33}$$

The leakage provides the a priori bound

$$|\eta(t)| \le \bar{\eta} := \max\left\{\eta_{0,\max}, \frac{\bar{e}_1}{\lambda_I}\right\}, \qquad \bar{e}_1 := \bar{x}_1 + \bar{y}_0, \tag{34}$$

with $\eta_{0,\max} := \max_{\mathcal{C}_0} |\eta(0)|$, since $|\eta(t)| \le e^{-\lambda_I t}|\eta(0)| + (1 - e^{-\lambda_I t})\bar{e}_1/\lambda_I$ is a convex combination of $|\eta(0)|$ and $\bar{e}_1/\lambda_I$, with $|e_1| \le \bar{e}_1$ by Proposition 1 and Assumption 2; a pure integrator admits no such bound. Any $\lambda_I > 0$ suffices for the analysis, and its value is chosen by simulation so as to preserve the integral action. The unsaturated observer-based ADRC command is

$$u_c = \frac{1}{\hat{b}}\left(\ddot{y}_d - \hat{\zeta} - k_1 e_1 - k_2\hat{e}_2 - k_I\eta\right), \tag{35}$$

where $k_1, k_2, k_I \in \mathbb{R}_{>0}$, and the applied voltage is $u = \mathrm{sat}_s(u_c)$, with sat_s defined in (14). The controller uses only the measured position x_1, the reference signals y_d, $\dot{y}_d$, $\ddot{y}_d$, and the observer states $\hat{x}_2$ and $\hat{\zeta}$. The variables x_2, $F(t)$, h, z, and the true input gain b are not used for feedback.

With ε bounded, the command and the saturation residual follow.

Lemma 5 (Command and saturation-residual bound)**.** *Let Assumptions 1–3 hold. Then, with* $\bar{\varepsilon}$ *from Proposition 2 and* $\bar{\eta}$ *from* (34)*,*

$$|u_c(t)| \le \bar{u}_c := \frac{1}{b_{\min}}\Big[\bar{y}_2 + \bar{\zeta}_0 + \omega_o^2\bar{\varepsilon} + k_1\bar{e}_1 + k_I\bar{\eta} + k_2(\bar{x}_2 + \bar{y}_1 + \omega_o\bar{\varepsilon})\Big] \tag{36}$$

and $|\Delta u(t)| \le |u_c(t)| \le \bar{u}_c$ *for all* $t \ge 0$*. The constant* $\bar{u}_c$ *combines independent worst cases and is used only to establish that the saturation residual is finite.*

Proof. See Section S-I of the supplementary material. □

Together, (15), (19), (33), and (35) define a dynamic output-feedback compensator with internal states $(\hat{x}_1, \hat{x}_2, \hat{\zeta}, \eta, \hat{b})$. The stability analysis below is carried out for this complete augmented closed-loop system.

Theorem 1. *(Adaptive saturated output-feedback ADRC tracking). Consider system* (3) *with output* (2)*. Let Assumptions 1–3 hold, and let* $\hat{b}(0) \in [b_{\min}, b_{\max}]$*. Fix compact sets* $\mathcal{X}_0 \subset \mathbb{R}^3$ *of plant initial conditions* $(x_1(0), x_2(0), z(0))$ *and* $\mathcal{C}_0$ *of compensator initial conditions* $(\hat{x}_1(0), \hat{x}_2(0), \hat{\zeta}(0), \eta(0))$*, and let* $\bar{x}_1, \bar{x}_2$ *be the constants of Proposition 1 made uniform over* $\mathcal{X}_0$ *as in Remark 1. Suppose the observer gains satisfy* (20) *for some* $\omega_o > 0$*, with no lower bound imposed on the observer bandwidth, let* $\lambda_I > 0$*, and let* k_1, k_2, k_I *satisfy* $k_I < k_1k_2$*. Then the scaled observer error* $\varepsilon = [\tilde{x}_1,\ \tilde{x}_2/\omega_o,\ \tilde{\zeta}/\omega_o^2]^\top$*, the tracking error* $\chi = [e_1, e_2, \eta]^\top$ *and the input-gain estimation error* $\tilde{b}$ *are uniformly bounded, and* (ε, χ) *is uniformly ultimately bounded, uniformly over* $\mathcal{X}_0 \times \mathcal{C}_0$*, over* $\hat{b}(0) \in [b_{\min}, b_{\max}]$ *and over* $\mathcal{F}$*.*

Proof. The observer error is already bounded by Proposition 2, so it enters below as a known input. The four steps build the tracking error dynamics, remove the unimplementable gain-mismatch term through the adaptive law, and close the loop with a Lyapunov argument.

Step 1: error variables. Define $\chi = [e_1, e_2, \eta]^\top$ (where $e_2 := x_2 - \dot{y}_d$) and $\tilde{b} := \hat{b} - b$, which satisfies the trivial bound (18) for all $t \ge 0$ regardless of γ_b. Since $\hat{e}_2 = e_2 + \tilde{x}_2$, the controller (35) couples χ to ε through $\tilde{x}_2 = \omega_o\varepsilon_2$. By Proposition 2, $\|\varepsilon(t)\| \le \bar{\varepsilon}$ for all $t \ge 0$, and by Lemma 5, $|\Delta u(t)| \le \bar{u}_c$.

Step 2: tracking error dynamics. Using $\dot{x}_2 = \hat{b}u + f_d(x_2) + \zeta - \hat{b}u$ from (25), $\hat{b}u = \hat{b}u_c + \hat{b}\Delta u$ and subtracting $\ddot{y}_d$,

$$\dot{\chi} = A_c\chi + B_c d_o + B_c f_d(x_2) - B_c\tilde{b}\,u + B_c\hat{b}\,\Delta u, \tag{37}$$

where $A_c := \begin{bmatrix} 0 & 1 & 0 \\ -k_1 & -k_2 & -k_I \\ 1 & 0 & -\lambda_I \end{bmatrix}$ has characteristic polynomial $s^3 + (k_2 + \lambda_I)s^2 + (k_1 + k_2\lambda_I)s + (k_I + k_1\lambda_I)$, which is Hurwitz whenever $k_I < k_2(k_1 + k_2\lambda_I + \lambda_I^2)$; the stated condition $k_I < k_1k_2$ is therefore sufficient for every $\lambda_I > 0$, $B_c = [0,1,0]^\top$, and, substituting $\hat{b}u_c = \ddot{y}_d - \hat{\zeta} - k_1e_1 - k_2\hat{e}_2 - k_I\eta$ and $\hat{e}_2 = e_2 + \tilde{x}_2$,

$$d_o = -k_2\tilde{x}_2 - \tilde{\zeta}, \qquad |d_o| \le (k_2\omega_o + \omega_o^2)\|\varepsilon\|. \tag{38}$$

Let $P_c \succ 0$ solve

$$A_c^\top P_c + P_cA_c = -I \tag{39}$$

and take as Lyapunov function candidate for the tracking error $V_c = \chi^\top P_c\chi$. Writing $x_2 = e_2 + \dot{y}_d$ and expanding x_2^3, Assumption 2 and Proposition 1 give $|e_2| \le \bar{e}_2 := \bar{x}_2 + \bar{y}_1$, so

$$f_d(x_2) = -\frac{c_p}{m_p}e_2^3 + \delta(t), \qquad |\delta(t)| \le \bar{\delta} := \frac{c_p}{m_p}\left(3\bar{e}_2^2\bar{y}_1 + 3\bar{e}_2\bar{y}_1^2 + \bar{y}_1^3\right), \tag{40}$$

and, decomposing $(P_c\chi)_2 = P_{c,21}e_1 + P_{c,22}e_2 + P_{c,23}\eta$,

$$2(P_c\chi)_2 f_d(x_2) \le -2\frac{c_p}{m_p}P_{c,22}e_2^4 + c_f\|\chi\| \le c_f\|\chi\| \quad (41)$$

with

$$c_f := 2\frac{c_p}{m_p}\bar{e}_2^3\sqrt{P_{c,21}^2 + P_{c,23}^2} + 2\|P_cB_c\|\,\bar{\delta},$$

obtained by Cauchy–Schwarz on $e_1e_2^3$, ηe_2^3 and $(P_c\chi)_2\delta$; the quartic term is non-positive (Lemma 2) and is discarded for a conservative bound.

Step 3: adaptive cancellation. The gain mismatch contributes $-2(P_c\chi)_2\tilde{b}u$ to $\dot{V}_c$, with $(P_c\chi)_2$ not implementable because e_2 is unmeasured. Augment the candidate with the parameter term $V_b := \tilde{b}^2/(2\gamma_b)$. By (15) and the projection property (17), $\dot{V}_b = \tilde{b}\,\mathrm{Proj}(\hat{b},\tau) \le \tilde{b}\tau = 2\tilde{b}u(P_c\hat{\chi})_2$, so that, using $\hat{e}_2 - e_2 = \tilde{x}_2$,

$$\begin{aligned} -2(P_c\chi)_2\tilde{b}u + \dot{V}_b &\le 2\tilde{b}u\big[(P_c\hat{\chi})_2 - (P_c\chi)_2\big] \\ &= 2\tilde{b}P_{c,22}\,u\,\tilde{x}_2 \le \bar{c}_\chi\,\omega_o\bar{\varepsilon}, \end{aligned} \quad (42)$$

with $\bar{c}_\chi := 2\bar{b}P_{c,22}u_{\max}$, by (18), $|u| \le u_{\max}$ and Proposition 2. The cancellation is exact up to the observer error, and the residual is a constant precisely because the regressor in (15) is the saturated voltage.

Step 4: tracking bound. With the composite Lyapunov function candidate $V_t := V_c + V_b$, and collecting (38), (41), (42), the saturation term $2(P_c\chi)_2\hat{b}\Delta u \le c_s\|\chi\|$ with $c_s := 2\|P_cB_c\|b_{\max}\bar{u}_c$ (Lemma 5), and $c_\omega := 2\|P_cB_c\|(k_2\omega_o + \omega_o^2)$,

$$\dot{V}_t \le -\|\chi\|^2 + \big(c_\omega\bar{\varepsilon} + c_f + c_s\big)\|\chi\| + \bar{c}_\chi\omega_o\bar{\varepsilon}. \quad (43)$$

Young's inequality on the linear term gives $\dot{V}_t \le -\frac{1}{2}\|\chi\|^2 + C_t$ with

$$C_t := \tfrac{1}{2}\big(c_\omega\bar{\varepsilon} + c_f + c_s\big)^2 + \bar{c}_\chi\omega_o\bar{\varepsilon}. \quad (44)$$

Since $V_b \le \bar{V}_b := \bar{b}^2/(2\gamma_b)$ by (18) and $\lambda_{\min}(P_c)\|\chi\|^2 \le V_c = V_t - V_b \le \lambda_{\max}(P_c)\|\chi\|^2$,

$$\dot{V}_t \le -\frac{1}{2\lambda_{\max}(P_c)}V_t + C_t + \frac{\bar{V}_b}{2\lambda_{\max}(P_c)}, \quad (45)$$

and the comparison lemma gives $\limsup_{t\to\infty} V_t \le 2\lambda_{\max}(P_c)C_t + \bar{V}_b$, hence

$$\limsup_{t\to\infty}\|\chi(t)\| \le \sqrt{\frac{2\lambda_{\max}(P_c)C_t + \bar{V}_b}{\lambda_{\min}(P_c)}}. \quad (46)$$

Together with Proposition 2 for ε and (18) for $\tilde{b}$, this bounds every error variable; the unscaled observer errors follow from $\tilde{x}_2 = \omega_o\varepsilon_2$ and $\tilde{\zeta} = \omega_o^2\varepsilon_3$. All the constants above depend only on fixed plant, reference and design data and on the compact-set bounds, so the ultimate bounds are uniform over $\mathcal{X}_0 \times \mathcal{C}_0$. Finally, since all closed-loop signals remain bounded on the maximal interval of existence, standard continuation results for projected dynamical systems exclude finite escape. This proves the theorem. □

Remark 3. *Theorem 1 establishes uniform ultimate boundedness, not exact asymptotic convergence. In particular, for a time-varying reference, the retained damping term generally satisfies $f_d(x_2) \to f_d(\dot{y}_d)$ when $e_2 \to 0$, which need not vanish if $\dot{y}_d \neq 0$.*

IV. Conclusions

An adaptive saturated output-feedback ADRC scheme was developed for a PEA with cubic viscous damping and Dahl hysteresis, using position measurements only. The tracking and observer errors are semiglobally uniformly ultimately bounded. The dose relation used in the supplementary material neglects fluid compliance and valve dynamics, which matter at the dosing tolerances typical of micropumps [15]. Future work will address experimental validation in a liquid medium and feedforward damping compensation to recover asymptotic tracking.

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